Yukyung Lee

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Data Science & Business Analytics Lab., Korea University
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RESEARCH INTERESTS

Writing with AI, Language Model Evaluation, Conversational AI, Anomaly Detection, Safety

EDUCATION

Korea University *Sept. 2019 - Present*
Ph.D. Candidate, Industrial and Management Engineering (Advisor: Prof. Pilsung Kang)

Hankuk University of Foreign Studies *Mar. 2015 - Aug. 2019*
Bachelor of Engineering, Industrial and Management Engineering (Advisor: Prof. Chungmok Lee)
Bachelor of Economics and Business, International Finance
· ***Summa Cum Laude***, Rank: 1st in the department

WORK EXPERIENCE

NAVER Search, Unsupervised Learning
· Role: Collaborative Researcher *Aug. 2023 - Present*
· Topic: AI-based writing, Writing evaluation

NAVER Search, Unsupervised Learning
· Role: Research Intern (Mentor: Soonwon Ka, Bokyung Son, Jaewook Kang) *Apr. 2023 - July 2023*
· Topic: Writing with Large Language Model
· Contributed to **CLOVA for Writing** @Naver

Upstage, NAVER Connect BoostCamp AI Tech *Jan. 2021 - Present*
· Generative AI Mentor (6th)
· Natural Language Processing Mentor (4th)
· Machine Reading Comprehension Mentor (2nd)
· Dialogue State Tracking Mentor (1st)

PUBLICATIONS

*J: Journal, C: Conference, W: Workshop, *: Equal cotribution*

[J5] RAPID: Training-free Retrieval-based Log Anomaly Detection with PLM considering Token-level information
Yukyung Lee*, Gunho No*, Hyeongwon Kang and Pilsung Kang.
Engineering Applications of Artificial Intelligence (Q1, IF=8) *(under 1st Review)*

[J4] KOreaPAS: TAPAS based Korean-Specific Table Question Answering
Suzie Oh, Yookyung Kho, Yukyung Lee and Pilsung Kang.
Journal of Korean Institute of Industrial Engineers, 2023

[W3] Painsight: An Extendable Opinion Mining Framework for Detecting Pain Points Based on Online Customer Reviews
Yukyung Lee, Jaehee Kim, Doyoon Kim, Yookyung Kho, Younsun Kim and Pilsung Kang.
ACL 2023 13th Workshop on Computational Approaches to Subjectivity, Sentiment & Social Media Analysis (Oral)

[J3, W2] DSTEA: Dialogue State Tracking with Entity Adaptive Pre-training
Yukyung Lee, Takyoung Kim, Hoonsang Yoon, Pilsung Kang, Junseong Bang, and Misuk Kim.
Knowledge-based System (Q1, IF=8.8) *(under 2nd Review)*
KDD 2023 Second Workshop on Knowledge Augmented Methods for NLP

[J2] LAnoBERT: System Log Anomaly Detection based on BERT Masked Language Model
Yukyung Lee, Jina Kim, Pilsung Kang.
Applied Soft Computing, 146, 110689, 2023 (Q1, IF=8.7)

[W1] Oh My Mistake!: Toward Realistic Dialogue State Tracking including Turnback Utterances
Takyoung Kim, Yukyung Lee, Hoonsang Yoon, Pilsung Kang, Junseong Bang, and Misuk Kim.
EMNLP 2022 Workshop of Towards Semi-Supervised and Reinforced Task-Oriented Dialog Systems

[C2] Mismatch between Multi-turn Dialogue and its Evaluation Metric in Dialogue State Tracking
Takyoung Kim, Hoonsang Yoon, Yukyung Lee, Pilsung Kang, and Misuk Kim.
Association for Computational Linguistics (ACL) 2022

[C1] Multi2OIE: Multilingual Open Information Extraction Based on Multi-Head Attention with BERT
Yongbin Ro, Yukyung Lee and Pilsung Kang.
Findings of ACL: EMNLP 2020

[J1] Drone Surveillance System Considering Dynamic POIs
Yukyung Lee, Munjeong Kang, Nayoung Kim, Chungmok Lee.
Journal of Korean Institute of Industrial Engineers, 45(5), 416-429, 2019

WORK IN PROCESS

Navigating the Path of Writing: Outline-guided Free-form Text Generation with Large Language Models
Yukyung Lee, Soonwon Ka, Bokyung Son, Jungsoo Park, Pilsung Kang, and Jaewook Kang.
Working paper

CheckEval: Checklist-based Framework for Robust LLM Evaluation with Human-Centric Alignment
Yukyung Lee, Joonghoon Kim, Jaehee Kim and Pilsung Kang.
Working paper

A Gradient Accumulation Methods for Contrastive Learning
Jaehee Kim, Yukyung Lee and Pilsung Kang.
Working paper

PROJECTS

Large Language Model Evaluation Method Discovery in Finance | *Granted by Kakaobank* *Oct. 2023 - present*
· Analyzed and benchmarked the performance of various LLMs tailored to the Korean financial sector to establish a comparative framework.
· Created a unified finance safety benchmark dataset and evaluation methodology, setting industry standards for the consistent and objective analysis of language models.

Table-based Machine Reading Comprehension | *Granted by Hanwha ICT* *June 2022 - Jan. 2023*
· Built TAPAS-based Korean TableMRC model and achieved state-of-the-art performance
· Conducted table data pre-training and collected text-table pair using dense retriever

Pain Point Detection with Customer Review | *Granted by LG Electronics* *Dec. 2021 - Apr. 2023*
· Proposed customer review analysis framework for detecting pain point
· Applied gradient-based saliency map to extract important topic clusters and keywords set

Dialogue State Tracking | *Granted by ETRI* *Apr. 2021 - Dec. 2022*
· Proposed selective knowledge masking for dialogue-specific pre-training
· Pointed out unrealistic assumptions of dataset and evaluation metric

Reliability Risk Suspected Process & Facility Search | *Granted by Samsung Electronics* *May 2020 - May 2021*
· Developed sequential pattern mining algorithm using transformer and VMSP
· Proposed reliability metrics for optimal facility sequence

Korean Summarization AI Dataset Construction | *Granted by NIA* *July 2020 - Dec. 2021*
· Developed abstractive summarization baseline model (KoPresumm)
· Released first human annotated Korean abstractive/extractive summarization dataset
· Organized Dacon summarization competition

Stock Market Prediction and Information Extraction with News | *Granted by SK C&C* *Aug. 2019 - Feb. 2020*
· Developed hierarchical attention network-based stock market prediction model from news text data
· Proposed BERT-based Korean information extraction system for analyzing key events affecting the stock price

Redefining Vehicle Delivery with Autonomous Cars via Reinforcement Learning *Sept. 2018 - Apr. 2019*
· Developed heuristic algorithms for the vehicle routing problem
· Proposed DQN-based optimization scheduling framework

Drone Surveillance System Considering Dynamic POIs *Jan. 2018 - June 2018*
· Developed real-time optimized drone route solver, Object detection engine
· Developed drone surveillance system (Application for drone auto-flight, Web for model serving)

AWARDS & HONORS

Factory Hack Korea (2020) | *KETI* 2nd Place
SKT Metatron Prize
· Machine ware prediction using manufacturing dataset

Kakao Arena (2019) | *Kakao* 3rd Place
· Article recommendation

INFORMS O.R & Analytics Student Team Competition (2019) | *General Motors* 3rd Place
· Outbound supply chain optimization using the delivery dataset

14th KIIE University Student Project Competition (2018) 1st Place
Minister's Prize of Science and ICT (과학기술정보통신부 장관상)
· Real-world Drone Surveillance framework considering the dynamic point of interest

KT&G Scholarship (2018) | *KT* -
Creative Innovation Leader for ICT-related student (1 year)

17th KOWIN Next-Generation Women Leaders' Idea Competition (2017) 1st Place
Minister's Prize of Gender Equality and Family (여성가족부 장관상)
· Proposed O2O Platform for women's voluntary education

Merit-based Scholarship (2017) | *Hankuk University of Foreign Studies* -
Half tuition and fees for undergraduate student (1 year)

PRESENTATIONS

Detecting System Log Anomalies in Real-Time using BERT without Additional Training *KIIE 2023*
Gunho No, Yukyung Lee and Pilsung Kang.

Customer Dissatisfaction Factor Detection based on Attribution Scores of Language Models using Customer Review Data *KIIE 2022*
Yukyung Lee, Hoonsang Yoon, Doyoon Kim, Yookyung Ko, Jaehee Kim, and Pilsung Kang.

A combined approach of reinforcement learning and mathematical programming for a real-world outbound supply chain optimization *KIIE 2019*
Munjeong Kang, Yukyung Lee, Jinyoung Cho, and Chungmok Lee.

A column generation approach for Drone Location and Routing with Altitude *KIIE 2018*
Yukyung Lee, Munjeong Kang, Nayoung Kim, and Chungmok Lee.

INVITED TALK

Task-Oriented Dialogue System and Korean Service Application *Mar. 2023*
MagicX

Deep Learning Based Natural Language Processing Models *Jan. 2023*
Department of Industrial & Data Engineering, Hongik University

Dialogue system model and evaluation *Nov. 2022*
Korea University Data Science Society, Korea University

TEACHING EXPERIENCE

Lecturer
- Natural Language Processing, Hankuk University of Foreign Studies (T02459101) *2024*
- Business Analytics, Korea University (IME654, Co-Lecturer) *2023*

Programming Instructor
- LG Energy Solution DX Expert Course (CNN, Transformer Lecture) *2022*
- LG Electronic DX Intensive Course (Text classification, Summzarization, Language Model Lecture) *2022*
- SK Hynix Data Analytics Course (Pytorch Basic, Deep Learning Lecture) *2021*
- HYUNDAI STEEL Data Analytics Course (Pytorch Basic, Deep Learning Lecture) *2021*

Teaching Assistant
- Fast Campus Advanced Machine Learning Course *2021*
- SK Hynix Data Analytics Course (Machine Learning, Deep Learning Lecture) *2020*
- LG Chem Expert Course (Machine Learning, Deep Learning Lecture) *2020*
- LG Innnotek Machine Learning Course (Python Basic Lecture) *2020*

Project Assistant
- Fast Campus Advanced Data Analytics Semi-professional Course *2022*
- Fast Campus Advanced Data Analytics Semi-professional Course *2021*

TECHNICAL SKILLS

Language: Python, R, HTML/CSS
Libraries: PyTorch, Tensorflow, Keras, Transformers, Pytorch-Lightning, Faiss, WandB, Google OR-Tools
Others: Git, Docker, LaTeX, Label-Studio

ACADEMIC SERVICE

Reviewer: NeurIPS dataset and benchmarks track (2022), Knowledge-Based System (2024), LREC-COLING (2024), ACL Rolling Review (2024)

REFERENCES

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Professor, HANKUK UNIVERSITY OF FOREIGN STUDIES
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